

CHEM&131: Introduction to Organic and Biochemistry

Fall 2026 (September 21 – December 10, 2026)

What's Covered

Everything covered in this course, organized by topic and module — useful for comparing against an equivalent course at another institution, and for previewing what's ahead before the term starts.

Activity numbers below are provisional (carried over from the Summer 2026 term) pending the Fall 2026 Chemfolio update — topics, labs, and objectives are current.

Module 1: Properties of Organic Molecules: Structure, IMF, and Naming

What this module covers:

- Distinguishing organic compounds from ionic inorganic compounds
- Common functional groups; shorthand structural drawing
- Acyclic alkanes, cycloalkanes, and constitutional isomers
- IUPAC naming of alkanes; predicting physical properties
- Complete and incomplete combustion of alkanes
- Lab: Lab Safety Orientation and Scavenger Hunt
- Required Reading: Smith Ch. 10 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: OM, CA33, CA34, CA35
- OOL1 Discussion
- ALEKS Problem Set: Ch. 10
- CLOs addressed: 1, 3 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day to drop online / 100% refund

Module 2: Properties of Organic Molecules: Functional Groups and Isomers

What this module covers:

- Unsaturated hydrocarbons: alkenes, alkynes, aromatic compounds; naming
- Constitutional isomers vs. stereoisomers; cis/trans isomers
- Saturated vs. unsaturated fatty acids and relative melting points
- Addition reactions of alkenes; hydrogenation; alkene-based polymers
- Alcohols, ethers, alkyl halides, thiols, aldehydes, ketones — naming and properties

- Alcohol dehydration/oxidation; thiol-to-disulfide conversion; aldehyde oxidation
- Chirality centers; Fischer projections
- Lab: Solubility
- Required Reading: Smith Ch. 11 (all), Ch. 12 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA36, CA37, CA38, CA39
- OOL2 Discussion
- ALEKS Problem Set: Ch. 11 and Ch. 12
- CLOs addressed: 1, 3 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day to add/drop without a "W"

Module 3: Properties of Organic Molecules: Functional Group Reactions

What this module covers:

- Characteristics and naming of carboxylic acids, esters, amines, amides
- Acid-base reactions of carboxylic acids; how soap cleans
- Condensation/hydrolysis reactions of carboxylic acids, esters, amides
- Acid-base reactions of amines; naming ammonium salts
- How penicillin works
- Lab: Consumer Plastics
- Required Reading: Smith Ch. 13 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA40, CA41, CA42, CA43
- OOL3 Discussion
- ALEKS Problem Set: Ch. 13
- CLOs addressed: 1, 2 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day for 50% refund
- Also this module: Quiz 1 (Smith Ch. 10 and 11)

Module 4: Proteins and Enzymes

What this module covers:

- Amino acid structure and acid-base properties
- N- and C-terminal labeling of peptides
- Primary, secondary, tertiary, quaternary protein structure
- Fibrous proteins (keratin, collagen); globular proteins (hemoglobin, myoglobin)
- Protein hydrolysis and denaturation
- Enzyme categories; diagnostic and therapeutic use of enzymes
- Lab: Chirality
- Required Reading: Smith Ch. 16 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA46B, CA46C, CA47A, CA47B
- OOL4 Discussion
- ALEKS Problem Set: Ch. 16
- CLOs addressed: 1, 2 (full text on the Course Learning Outcomes page)
- Calendar note: Friday — last day to withdraw for the quarter
- Also this module: Quiz 2 (Smith Ch. 12 and 13)

Module 5: Carbohydrates

What this module covers:

- Three major carbohydrate types; monosaccharide structure
- Cyclic forms of monosaccharides; α/β classification
- Oxidation/reduction products of monosaccharides
- Cellulose, starch, and glycogen; carbohydrates and blood type
- Lab: Reactions
- Required Reading: Smith Ch. 14 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA44B, CA44C, CA50, CA53
- OOL5 Discussion
- ALEKS Problem Set: Ch. 14
- CLOs addressed: 1, 2 (full text on the Course Learning Outcomes page)

Module 6: Lipids

What this module covers:

- General characteristics of lipids
- Fatty acid classification and melting point relationships
- Wax, triacylglycerol, and phospholipid structure; fat vs. oil; hydrolysis products
- Cell membrane structure and transport mechanisms
- Steroid structure (cholesterol) and steroid hormones
- Fat-soluble vitamins
- Lab: Proteins
- Required Reading: Smith Ch. 15 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA45B, CA45C, CA45D, CA54A
- OOL6 Discussion
- ALEKS Problem Set: Ch. 15
- CLOs addressed: 1, 2 (full text on the Course Learning Outcomes page)

Module 7: Nucleic Acids

What this module covers:

- Nucleoside and nucleotide structure; drawing DNA/RNA segments
- DNA double helix; steps of replication
- RNA types and functions; transcription and translation
- Genetic code; mutation and genetic disease
- DNA fingerprinting; characteristics of viruses
- Lab: Sugar Spill
- Required Reading: Smith Ch. 17 (all)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA48A, CA48B, CA48C
- OOL7 Discussion
- ALEKS Problem Set: Ch. 17
- CLOs addressed: 1, 2, 3 (full text on the Course Learning Outcomes page)

Module 8: Metabolism — Part 1

What this module covers:

- Definition of metabolism; four stages of catabolism
- Role of ATP in energy production
- Main coenzymes used in metabolism
- Required Reading: Smith Ch. 18 (part 1 of 2 — continues into Module 9)

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA49A, CA49B, CA49C
- OOL8 Discussion
- CLOs addressed: 1, 2, 3 (full text on the Course Learning Outcomes page)
- Also this module: Quiz 4 (Smith Ch. 16 and 17)

Module 9: Metabolism — Part 2

What this module covers:

- Glycolysis; pathways for pyruvate metabolism
- Citric acid cycle
- Electron transport chain and oxidative phosphorylation
- Energy yield from glucose metabolism
- β -oxidation of fatty acids; ketone body structure and role
- Lab: Seed Toxicology
- Required Reading: Smith Ch. 18 (part 2 of 2)
- Final Project: Teach Your Community (TYC) due

▼ Course outcomes & weekly logistics

- 3-2-1 TEAM Activities: CA51, CA52, CA55, CA56
- OOL9 + OOL10 Discussions
- ALEKS Problem Set: Ch. 18
- CLOs addressed: 1, 2, 3 (full text on the Course Learning Outcomes page)
- Also this module: Quiz 5 (Smith Ch. 18)